

Digital Forensics Final Exam

Question 1

The names, aliases, and descriptions presented in this content are entirely fictitious and created solely for illustrative purposes. Any resemblance to actual persons, living or deceased, businesses, organizations, or real-life events is purely coincidental. This material is not intended to imply or suggest any real-world association or involvement in unlawful activities.

You are part of a **digital forensics team** tasked with investigating a disk image (finalNTFS.dd) obtained from a suspect's computer. The disk contains multiple partitions, and some files appear to be encrypted using information stored on other partitions. This disk image was acquired during an investigation into suspected **cybercriminal activities**.

Use **Autopsy** to work on this question.

Part 1: Chat Log Recovery

We identified that the **first partition** contains a **chat log** between the suspects. Your first task is to recover this file and provide answers to the questions below to help us understand the nature of this criminal activity:

- **Q1:** What are the aliases of the suspects mentioned in the chat log?
- **Q2:** What roles have the suspects assigned to themselves for the operation?
- **Q3:** What specific tasks or objectives are outlined in the chat?
- **Q4:** Are there references to specific files, tools, or locations critical to the operation?
- **Q5:** How does the decision to “go dark” on the communication channel after the meeting impact the forensic investigation?

Part 2: File Recovery on the Second Partition

After reviewing the chat log, we learned that **second partition** contains additional files suspected to be related to the criminal operation. Your task is to **recover these files** and identify their relevance to the investigation. Document the recovery process and note any encrypted or obfuscated files.

Part 3: Suspect Identification

It has been determined that the suspects are using **aliases**. Your task is to recover additional forensic information from the disk to identify and actual names the suspects. Specifically:

- **Q1:** What is the real name of their leader?
- **Q2:** Provide a comprehensive list of suspects from the chat log, including:
 - Alias
 - Real Name
 - Suspect ID
 - Role in the crime

Part 4: Headshot Recovery

We have reason to believe that the suspects stored their **headshots** on the disk. Your task is to recover all files containing headshots of individuals involved in this crime. For each recovered file, you must:

- Generate **MD5** and **SHA256** hash values to ensure file integrity.
- Recover and verify the files with their correct extensions using the file signatures.
- Document the recovered headshots with their corresponding hash values and their actual extensions.
- Fore each file, document whether the file was available or was deleted.

Deliverables

1. Detailed Report:

- Answers to all questions/items outlined in Parts 1–4.
- Explanation of the methods used to recover each file.
- Your report has to be properly organized and labeled to reflect the parts listed in the question (use the given template to provide your responses).

2. Annotated Screenshots:

1. Provide evidence supporting your findings, including annotated screenshots of your recovery process, file contents and verification process.

3. A copy of all recovered files should be included in the question folder (use the provided template). All recovered files should be accessible without a password.
4. Make sure that all your files are properly organized and labeled.

Question 2

You've been given a file named "MysteryMemory.zip," which contains an image of a system's memory dump. Your job is to analyze this artifact and extract specific pieces of information.

Please carry out the following analyses:

1. Identify the image information.
2. Compile a List of security identifiers and associated processes.
3. Extract users and system credential (hashes).

Documentation

- Execute the necessary commands in your terminal or command prompt to find this information.
- Capture and provide a **Screenshot** of your terminal or command prompt, clearly showing the commands used and their output.

Question 3

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Network Forensics Investigation

Objective:

Your task is to analyze network traffic logs to investigate a potential data breach. Using captured network traffic files, you will sequentially examine DNS, HTTP, and FTP protocols

to answer key forensic questions about the incident. Your goal is to identify the malicious activities, reconstruct the sequence of events, and recover evidence.

Scenario:

A confidential employee list file has been leaked. An investigation reveals that a malicious website was accessed to obtain this file. Additionally, subsequent activity on an FTP server was detected, involving sensitive files and instructions.

You are provided with network traffic logs (FinalCapture.pcap) that include DNS, HTTP, and FTP traffic.

Your Tasks:

Part 1: DNS Analysis

1. Identify the **domain name** of the malicious website visited.
2. Find the **IP address** of the website resolved via DNS.
3. Find the **IP and MAC** address of the device used to make the DNS query

Part 2: HTTP Analysis

Using HTTP protocol analysis, answer the following:

1. Determine the **operating system (OS)** of the client device.
2. Find the **full URL** used to download the confidential employee list file.
3. Identify the **timestamp** when the download of the employee list file was successfully completed.
4. List the **employee details** found in the confidential employee list file.

Part 3: FTP Analysis

Using FTP protocol analysis, answer the following:

1. Identify all the **user credentials (username)** used to log in and the **timestamp of successful login** to the FTP server.
2. Identify the **timestamp** at which the README.txt file was successfully downloaded.
3. Examine and share a screenshot of the **txt file**.
4. Recover any subsequent files found in the FTP traffic.

5. Provide the screenshot of **final file** You will have to perform decryption if necessary. Provide the hashes of decrypted and encrypted files (if a file is encrypted).

Deliverables:

Report Document:

- A written report answering all the above questions. Include annotated screenshots from your analysis tool (e.g., Wireshark) to support your findings (Use the provided template)

Recovered Evidence:

Submit the following recovered files in the corresponding folder (use the template):

1. Screenshot of the confidential employee list file.
2. The README.txt file.
3. Any subsequent files obtained through the FTP analysis.

Question 4

You have been provided with a zip file titled "Windows Registry," which contains various registry artifacts from a Windows 10 user's system. Your task is to analyze these artifacts to extract specific pieces of information. Please carry out the following analyses:

A. Compile a **List of User Accounts** present on the system.

B. Create a table displaying the **Name of the Last Accessed File** and the corresponding **Time of Access**.

Documentation:

- Execute the necessary commands in your terminal or command prompt to find this information.
- Capture and provide a **Screenshot** of your terminal or command prompt, clearly showing the commands used and their output.

Question 5

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The team at a regional transmission operator, **East River Power Authority (ERPA)**, has developed their weekly operations schedule for describing peak load windows, maintenance outages, EMS/SCADA backup times, and remote access rules. This is a critical information and must be kept confidential and not be accessed by anyone outside the operator personal. Anyone with that schedule can identify hours with the weakest operational margin and the best windows for undetected changes to relays and configurations.

Yesterday, ERPA's security team detected suspicious traffic from a workstation assigned to a junior engineer, **Dustin Henderson**, who had access to the schedule as part of his responsibilities. Network monitoring flagged repeated uploads of harmless-looking media files from **Dustin's** workstation to an external file sharing site during off-hours.

Internal investigators cloned **Dustin's** workstation and seized a collection of files from his workstation.

You are given a forensic copy of the files from the suspect's workstation, all in a single Zip format file (DiskImage-DustinHenderson.zip). While the files appear normal, you will need your forensics skills to verify and determine if the operational schedule has been compromised. You must determine:

Part 1: Whether any of the provided files contain hidden data

Part 2: Whether any of the files have protected info? If so, how the password scheme works?

Part 3: Whether the recovered content is indeed sensitive power system material?